

MODEL EXIT EXAM 2026 — QUESTIONS & ANSWERS

Q1. In a CIDR block of 192.168.100.0/23, what is the broadcast address?

- a. 192.168.102.0
 - b. 192.168.100.1
 - c. 192.168.100.255
 - d. 192.168.101.255
-

Q2. Which of the following describes the possible alternative ways in which files on a disk can be accessed?

- a. serial or random
 - b. object or serial
 - c. text or binary
 - d. binary or random
 - e. random or binary
-

Q3. In Agile development, what is a sprint?

- a. A testing phase
 - b. A debugging session
 - c. A time-boxed development cycle
 - d. A rapid design process
-

Q4. Decision trees classify data points by recursively splitting the data space based on _____ features.

- a. numerical
 - b. most relevant
 - c. most correlated
 - d. all available
-

Q5. What is meant by software quality in software engineering?

- a. Complexity of code
- b. High number of features
- c. Degree to which software meets requirements and is free of defects
- d. Low cost

Q6. Which class of IP addresses is reserved for experimental use?

- a. Class A
- b. Class E
- c. Class C
- d. Class D

Q7. Which of the following search algorithms guarantees both completeness and optimality, assuming the heuristic is admissible and consistent?

- a. Greedy Best-First Search
- b. A* Search
- c. Depth-First Search
- d. Hill Climbing

Q8. Which protocol uses port 443 and operates at Layer 7?

- a. FTP
- b. HTTPS
- c. SSH
- d. SNMP

Q9. Consider the following method, setValue, which adds a value into an array list: `public void setValue(int position, String valueIn) { list[position-1] = valueIn; }` Assuming that list has been declared as an array of Strings, which one of the following might cause the setValue method to throw an `ArrayIndexOutOfBoundsException` exception:

- a. the type of the parameter position being declared as an int
- b. not declaring the method static
- c. the return type being declared as void
- d. the value of the first parameter

Q10. Which Activity lifecycle callback is the first one called after the Activity is launched and is where initial setup like inflating the UI layout typically occurs?

- a. `onStart()`
 - b. `onCreate()`
 - c. `onResume()`
-

d. onAttach()

Q11. Which CSS property is used to control the space between the content and the border of an element?

- a. padding
 - b. margin
 - c. spacing
 - d. border-spacing
-

Q12. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

- a. Entity set
 - b. Relation set
 - c. Tuples
 - d. Entity Relation model
-

Q13. In the TCP/IP model, which layer corresponds to the Transport layer of OSI?

- a. Transport
 - b. Application
 - c. Network Access
 - d. Internet
-

Q14. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation. What integrity constraint dictates this?

- a. Referencing
 - b. Referential
 - c. Specific
 - d. Primary
-

Q15. Which of the following is not a Boolean expression?

- a. $10 == 2*5$
- b. $20 = 5+15$
- c. $10**2 != 100$

d. True != True

Q16. Which of the following data structures uses the LIFO (Last In First Out) principle?

- a. Binary Tree
 - b. Stack
 - c. Linked List
 - d. Queue
-

Q17. What will be the output of the following code? `var x = 5; var y = '5'; console.log(x == y);`

- a. true
 - b. undefined
 - c. null
 - d. false
-

Q18. Consider the following method, `setValue` (same as Q9). Assuming that `list` has been declared as an array of Strings, which one of the following might cause the `setValue` method to throw an `ArrayIndexOutOfBoundsException` exception:

- a. not declaring the method static
 - b. the type of the parameter position being declared as an int
 - c. the value of the first parameter
 - d. the method having more than one parameter
 - e. the return type being declared as void
-

Q19. What will be printed when the following code is executed? `>>> tuple1 = (1, 2, 3, 4) >>> print(tuple1[1:-1])`

- a. (2, 3)
 - b. Error: tuple slicing
 - c. (2, 3, 4)
 - d. (1, 2, 3, 4)
-

Q20. In COCOMO model, Initial estimate E_i = _____

- a. $a * (KLOC) * b$
- b. $b * (KLOC) * a$

- c. a(KLOC)b
- d. b(KLOC)a

Q21. A process goes from running-state to blocked-state whenever a running process:

- a. moved to secondary storage because of memory shortage
- b. None
- c. interrupted by a higher priority process
- d. CPU executes context switching
- e. needs to get input/output in the middle of execution

Q22. Which of the following is not true about functions in python?

- a. Functions are reusable pieces of program
- b. Functions provide better modularity for your programs
- c. Function definitions can alter flow of execution
- d. Python allows user defined functions

Q23. Which Android UI component is described as a 'microactivity' that represents a behavior or portion of the UI and must be hosted within an Activity?

- a. View
- b. Adapter
- c. Fragment
- d. Service

Q24. Which method is used to add an element to the end of an array in JavaScript?

- a. pop()
- b. push()
- c. shift()
- d. unshift()

Q25. Four conditions are required for a deadlock to occur. Which one of the following is NOT a condition of deadlock?

- a. Circular wait

- b. Hold and wait
- c. Mutual exclusion
- d. No preemption
- e. Busy wait

Q26. Mutual execution algorithms are primarily used to resolve:

- a. None
- b. Busy-waiting
- c. Deadlock
- d. Starvation
- e. Race condition

Q27. HMMs are useful for modeling sequential data where the underlying states are ____.

- a. Pre-defined
- b. partially observable
- c. always observable
- d. completely random

Q28. What will be printed when the following code is executed? >>> tuple1 = (1, 2, 3, 4) >>> print(tuple1[1:-1]) (same as Q19)

- a. (2, 3)
- b. Error: tuple slicing
- c. (2, 3, 4)
- d. (1, 2, 3, 4)

Q29. In COCOMO model, Initial estimate E_i = _____ (same as Q20)

- a. $a \cdot (\text{KLOC})^b$
- b. $b \cdot (\text{KLOC})^a$
- c. $a(\text{KLOC})b$
- d. $b(\text{KLOC})a$

Q30. Which of the following is true regarding the flow of execution in a while statement?

- a. If the condition is yielding not true, execute the body
 - b. If the condition is false, exit the while statement and continue execution
 - c. All
 - d. Evaluates the condition yielding only True
-

Q31. If an algorithm has a time complexity of $O(n^2)$, what happens to the running time if the input size is doubled?

- a. It triples
 - b. It doubles
 - c. It quadruples
 - d. It remains the same
-

Q32. Which one of the following scheduling algorithms most likely to cause starvation?

- a. Round-Robin (RR)
 - b. First-Come, First Served (FCFS)
 - c. Short-Job First (SJF)
 - d. None
 - e. Priority-Based With Aging
-

Q33. Bereket needs to send an email to his supervisor with an attachment that includes sensitive information. He wants to maintain the confidentiality of this information. Which of the following choices is the BEST choice to meet his needs?

- a. Digital signature
 - b. Encryption
 - c. Hashing
 - d. Data masking
-

Q34. The main purpose of STP (Spanning Tree Protocol) is to:

- a. Prioritize traffic
 - b. Monitor SNMP traps
 - c. Prevent Layer 2 loops
 - d. Filter traffic by VLAN
-

Q35. Consider the following setValue method (same as Q9/Q18). Which one might cause an ArrayIndexOutOfBoundsException exception:

- a. not declaring the method static
 - b. the type of the parameter position being declared as an int
 - c. the value of the first parameter
 - d. the method having more than one parameter
 - e. the return type being declared as void
-

Q36. Which command can display the current ARP cache on a Windows system?

- a. ipconfig
 - b. ping
 - c. arp -a
 - d. netstat -r
-

Q37. In a graph, what does a cycle represent?

- a. A linear path
 - b. A spanning tree
 - c. A set of disconnected vertices
 - d. A path that starts and ends at the same vertex
-

Q38. Which of the following is a divide and conquer algorithm?

- a. Binary Search
 - b. Insertion Sort
 - c. Linear Search
 - d. Bubble Sort
-

Q39. Which of the following is true regarding the flow of execution in a while statement? (same as Q30)

- a. If the condition is yielding not true, execute the body
 - b. If the condition is false, exit the while statement and continue execution
 - c. All
 - d. Evaluates the condition yielding only True
-

Q40. Which one of the following scheduling algorithms most likely to cause starvation? (same as Q32)

- a. Round-Robin (RR)
- b. First-Come, First Served (FCFS)
- c. Short-Job First (SJF)
- d. None
- e. Priority-Based With Aging

Q41. Which one of the following scheduling algorithms most likely to cause starvation? (same as Q32)

- a. Round-Robin (RR)
- b. First-Come, First Served (FCFS)
- c. Short-Job First (SJF)
- d. None
- e. Priority-Based With Aging

Q42. Bereket needs to send an email to his supervisor with an attachment that includes sensitive information. (same as Q33)

- a. Digital signature
- b. Encryption
- c. Hashing
- d. Data masking

Q43. Consider the following setValue method — which one might cause an ArrayIndexOutOfBoundsException exception:

- a. not declaring the method static
- b. the type of the parameter position being declared as an int
- c. the value of the first parameter
- d. the method having more than one parameter
- e. the return type being declared as void

Q44. Which of the following best describes a compiled language?

- a. Code is always faster in compiled languages because they have no errors
- b. Code is executed line-by-line
- c. Code is translated directly into machine code before execution

d. Compiled languages cannot be used for low-level programming

Q45. Which principle refers to minimizing the interdependence between software modules?

- a. Modularity
 - b. Cohesion
 - c. Abstraction
 - d. Coupling
-

Q46. _____ is a set of one or more attributes taken collectively to uniquely identify a record.

- a. Primary Key
 - b. Foreign key
 - c. Super key
 - d. Candidate key
-

Q47. Consider the inheritance hierarchy shown in the diagram below: [Class inheriting diagram with two subclasses]. What does the arrow from subclasses to Class represent?

- a. Inheritance
 - b. Composition
 - c. Aggregation
 - d. Association
-

Q48. You are comparing different types of authentications. Of the following choices, which one uses multifactor authentication?

- a. A system that requires users to enter a username and password
 - b. A system that requires users to have a smart card and a PIN
 - c. A system that checks an employee's fingerprint and does a vein scan
 - d. A cipher door lock that requires employees to enter a code to open the door
-

Q49. Which evaluation metric is most suitable for imbalanced classification problems?

- a. Accuracy
- b. R-squared
- c. F1-Score

d. Mean Squared Error (MSE)

Q50. In linear regression, the predicted value is a linear function of the ____.

- a. distance from the hyperplane
 - b. independent variables
 - c. class probabilities
 - d. dependent variable
-

Q51. Your organization hired a cybersecurity expert to perform a security assessment. After running a vulnerability scan, she sees: Host IP 192.168.1.10 OS Apache httpd 2.4.33 Vulnerable to mod_auth exploit. However, she verified that the mod_auth module has not been installed or enabled on the server. Which of the following BEST explains this scenario?

- a. A false positive
 - b. The result of a credentialed scan
 - c. The result of a non-credentialed scan
 - d. A false negative
-

Q52. What is the main function of the three-way handshake in TCP?

- a. Flow control
 - b. Authentication
 - c. Reliable data delivery
 - d. Connection setup and synchronization
-

Q53. A class called Network has been declared with private attributes maxNumberOfUsers and currentNumberOfUsers. A class SubNet extends Network. In a program that declares and initializes an object sub of SubNet, which of the following statements would NOT cause a compiler error?

- a. `sub.maxNumberOfUsers = 100;`
 - b. `sub.setMaxNumberOfUsers(100);`
 - c. `sub.currentNumberOfUsers = 50;`
 - d. `System.out.println(sub.maxNumberOfUsers);`
-

Q54. Which one of the following mechanisms allows execution of multiple processes by switching CPU even if the currently executing process is not completed?

- a. Time-sharing

- b. Multiprogramming
- c. Multiprocessing
- d. None
- e. Monoprogramming

Q55. _____ involves generating, collecting, disseminating, and storing information.

- a. Concurrent Management
- b. Critical Management
- c. Configuration Management
- d. Communication Management

Q56. Chala is a security practitioner tasked with ensuring that the information on the organization's public website is not changed by anyone outside the organization. This task is an example of ensuring _____.

- a. Availability
- b. Integrity
- c. Confidentiality
- d. Authentication

Q57. Which software process model emphasizes strict sequential flow between phases?

- a. Agile Model
- b. Incremental Model
- c. Spiral Model
- d. Waterfall Model

Q58. What is the output of the following code? >>> my_list = [1,2,3,2,4] >>> my_list.remove(2) >>> my_list

- a. None
- b. [1,2,2,4]
- c. [1,3,2,4]
- d. [1,3,4]

Q59. Which of the following best describes the role of a file system in an operating system?

- a. It manages the computer's hardware resources, such as CPU and memory

- b. It provides a structured way to store, organize, retrieve, and manage files on storage devices
- c. It optimizes network bandwidth usage for file transfers over the internet
- d. It handles user authentication and access control for logging into the system
- e. None

Q60. Delphi approach is also called _____ group consensus technique.

- a. Fully Consultative
- b. Non-Consultative
- c. Consultative
- d. Partially Consultative

Q61. What is the output of the following code? >>> my_list = [1,2,3,4,5,6,7,8,9,10] >>> my_list[5::-2]

- a. [6,4,2]
- b. [5,7,9]
- c. [5,4,3,2,1]
- d. []

Q62. _____ is a hierarchical and incremental decomposition of the project into phases, deliverables and work packages.

- a. Work Integration Structure
- b. Task Integration Structure
- c. Bottom-up Integration Structure
- d. Work Breakdown Structure

Q63. The bias-variance trade-off implies that:

- a. Increasing model complexity reduces both bias and variance
- b. Underfitting is caused by low variance
- c. Simple models have high bias and low variance
- d. Overfitting occurs when bias is too high

Q64. What is the primary objective of virtual memory in a computer system?

- a. increase the physical memory capacity

- b. None
 - c. increase execution speed by catching frequently used data
 - d. allow processes use more memory than physical available
 - e. secure part of memory from being accessed by other processes
-

Q65. What is printed by the following statements? >>> str = 'This is python code' >>> print(str[2] + str[-6])

- a. tn
 - b. io
 - c. in
 - d. to
-

Q66. Sara is a database manager. Sara is allowed to add new users to the database, remove current users, and create new usage functions for the users. Sara is not allowed to read the data in the fields of the database itself. This is an example of:

- a. Role Based Access Control
 - b. Discretionary Access Control
 - c. Mandatory Access Control
 - d. Behavior Based Access Control
-

Q67. Which type of data can be stored in the database?

- a. Text, files containing data
 - b. All of the above
 - c. Image oriented data
 - d. Data in the form of audio or video
-

Q68. What is the primary function of an Intent in Android app navigation?

- a. To manage background threads for long-running operations
 - b. To request an action from another app component
 - c. To store application settings persistently
 - d. To define the visual appearance of a screen
-

Q69. What is the primary purpose of WAI-ARIA?

- a. To improve website SEO rankings
 - b. To make web applications more accessible
 - c. To increase website loading speed
 - d. To replace HTML with a new accessibility standard
-

Q70. A program is required to add two integer numbers together. The GUI for this program shows two text fields and a Calculate button. What is the component labeled with an arrow in the diagram?

- a. JFrame
 - b. JTextField
 - c. JLabel
 - d. JButton
-

Q71. What is meant by software quality in software engineering? (same as Q5)

- a. Complexity of code
 - b. High number of features
 - c. Degree to which software meets requirements and is free of defects
 - d. Low cost
-

Q72. The specific set of sequential tasks upon which the project completion date depends or the longest full path is called ____.

- a. Full path
 - b. Project Path
 - c. Critical Path
 - d. Complete Path
-

Q73. Which of the following sorting algorithms has the best average-case time complexity?

- a. Quick Sort
 - b. Bubble Sort
 - c. Insertion Sort
 - d. Selection Sort
-

Q74. What is the worst-case time complexity of Merge Sort?

- a. $O(n^2)$
 - b. $O(\log n)$
 - c. $O(n \log n)$
 - d. $O(n)$
-

Q75. How do you check if an element exists in a set? Using

- a. in keyword
 - b. exists()
 - c. contains()
 - d. check()
-

Q76. Which of the following is an example of a 'Something you know' authentication factor?

- a. GPS
 - b. Finger Printing
 - c. Password
 - d. ATM Card
-

Q77. What is the output of this expression: `>>> (2 + 5) ** 2 // 10 + 2 * 3`

- a. 12.0
 - b. 10
 - c. 8
 - d. 12
-

Q78. A technique to prevent modification anomalies by creating a different class of relations in RDMS are called?

- a. Functional Dependencies
 - b. Normal Forms
 - c. Referential integrity
 - d. Data integrity
-

Q79. Which operation is used to extract specified columns from a table?

- a. Substitute

- b. Extract
- c. Project
- d. Join

Q80. A team is designing a weather app that must work offline in remote areas but also sync data when internet connectivity is available. Which architectural approach addresses this need?

- a. Only online-based architecture
- b. Peer-to-peer architecture
- c. Client-side caching with periodic synchronization
- d. Pure serverless architecture (no offline support)

Q81. Which of the following is the primary goal of software engineering?

- a. To produce reliable, efficient, and maintainable software
- b. To develop low-level machine code
- c. To increase hardware performance
- d. To create visually appealing user interfaces

Q82. Which of the following is a divide and conquer algorithm? (same as Q38)

- a. Binary Search
- b. Insertion Sort
- c. Linear Search
- d. Bubble Sort

Q83. In the context of search problems, what does a heuristic function do?

- a. Estimates the cost from a node to the goal
- b. Calculates the actual cost of a path
- c. Calculates the cost from start to goal
- d. Finds the goal state directly

Q84. How do you create a CSS rule that applies only to screen devices with a maximum width of 600px?

- a. @media screen and (min-width: 600px) { ... }
- b. @media print screen and (max-width: 600px) { ... }

- c. @media screen and (max-width: 600px) { ... }
 - d. @media all and (max-width: 600px) { ... }
-

Q85. What is the main advantage of using kernel methods in machine learning?

- a. They are faster than other machine learning algorithms for all tasks
 - b. They can only be used with Support Vector Machines (SVMs)
 - c. They require less data for training compared to other methods
 - d. They allow linear algorithms to work effectively with non-linear data
-

Q86. Which CSS property is used to control the stacking order of positioned elements?

- a. display
 - b. position
 - c. float
 - d. z-index
-

Q87. How can web developers accommodate users with mobility impairments?

- a. Ensuring all controls are accessible via keyboard
 - b. Making users rely on a mouse for navigation
 - c. Limiting access to only touch-screen devices
 - d. Removing all interactive elements
-

Q88. Consider class ClassA (with testMethod printing 'Hello') and class ClassB extends ClassA (with testMethod printing 'Goodbye'). If a ClassB object calls testMethod(), what is printed?

- a. Hello
 - b. Goodbye
 - c. Hello then Goodbye
 - d. Goodbye then Hello
-

Q89. Your organization hired a cybersecurity expert. After a vulnerability scan: Host IP 192.168.1.10 OS Apache httpd 2.4.33 Vulnerable to mod_auth exploit. She verified that mod_auth has NOT been installed. Which BEST explains this scenario?

- a. The result of a credentialed scan
- b. A false negative

- c. The result of a non-credentialed scan
- d. A false positive

Q90. What is printed by the following statements? >>> Str1 = 'PYTHON programming' >>> print(Str1[2] * Str1.index('O'))

- a. OOOO
- b. THON
- c. TTTT
- d. YYYY

Q91. What is the output of the following code? >>> my_list = [1,2,3,4,5,6,7,8,9,10] >>> my_list[5::-2] (same as Q61)

- a. [6,4,2]
- b. [5,7,9]
- c. [5,4,3,2,1]
- d. []

Q92. Which statement about ensemble methods (e.g., Random Forests) is TRUE?

- a. They require all base models to be identical
- b. They perform poorly compared to single models
- c. They combine weak learners to reduce bias only
- d. They average predictions to decrease variance

Q93. What is the primary purpose of the RecyclerView widget?

- a. To display a single image on the screen
- b. To display lists of data by efficiently reusing item views
- c. To provide a simple button with click functionality
- d. To manage background tasks efficiently

Q94. When you want to navigate to a specific Activity within your application, which type of Intent is typically used?

- a. Explicit intent
- b. Implicit intent

- c. Pending intent
 - d. Broadcast intent
-

Q95. What is the purpose of the box-sizing property in CSS?

- a. To include padding and border in the element's total size
 - b. To set the total margin of the box
 - c. To define the size of the total content and margin box
 - d. To apply a shadow and margin to the box
-

Q96. Your organization's backup policy for a file server dictates that the amount of time needed to restore backups should be minimized. Which of the following backup plans would BEST meet this need?

- a. Full backups on Sunday and incremental backups on the other six days of the week
 - b. Full backups on Sunday and differential backups on the other six days of the week
 - c. Differential backups on Sunday and incremental backups on the other six days of the week
 - d. Incremental backups on Sunday and differential backups on the other six days of the week
-

Q97. In a neural network, the activation function introduces:

- a. Regularization to prevent overfitting
 - b. Non-linearity to model complex patterns
 - c. Dimensionality reduction
 - d. Linear decision boundaries
-

Q98. A _____ is the lowest level of work on the project.

- a. Task Set
 - b. Task
 - c. Milestone
 - d. Work product
-

Q99. Return of Investment (ROI) ROI = _____

- a. $(\text{Gain} - \text{Cost}) / \text{Gain}$
- b. $(\text{Cost} - \text{Gain}) / \text{Gain}$
- c. $(\text{Gain} - \text{Cost}) / \text{Cost}$

d. $(\text{Cost} - \text{Gain}) / \text{Cost}$

Q100. A startup is building an e-commerce platform expecting rapid growth. They need an architecture that allows independent scaling of the product catalog, payment processing, and recommendation engine. Which architectural style is most suitable?

- a. Single-tier architecture (all logic on the client side)
 - b. Monolithic architecture (all components in one codebase)
 - c. Shared-database architecture (all services use one database)
 - d. Microservices architecture (independent, loosely coupled services)
-

Q101. Slack = _____

- a. LS-EF
 - b. ES-LF
 - c. ES-LS
 - d. LF-EF
-

Q102. Which of the following is the correct syntax to create a function in JavaScript?

- a. function: myFunction() {}
 - b. function = myFunction() {}
 - c. function => myFunction() {}
 - d. function myFunction() {}
-

Q103. Which of the following function declarations is correct to define a function that accepts a variable number of arguments?

- a. def func(*args):
 - b. Not possible in python
 - c. def func(args*):
 - d. def func(**args):
-

Q104. Which of the following is a key challenge in reinforcement learning?

- a. Requirement of perfectly balanced datasets
- b. The trade-off between exploration and exploitation
- c. Lack of labeled training data

d. High computational cost of matrix operations

Q105. Which algorithm is most appropriate when the cost of actions varies and finding the least-cost solution is important?

- a. Depth-First Search
 - b. Breadth-First Search
 - c. Greedy Best-First Search
 - d. Uniform Cost Search
-

Q106. How can you apply a CSS Grid layout to a container?

- a. display: block;
 - b. display: grid;
 - c. display: inline-grid;
 - d. display: flex;
-